Answer key: Lesson 9, Picture Equations

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.

Answer: 8

11 - 3 = 8, so the cup holds 8 counters. Check: 8 + 3 = 11.

Watch for: Writing 14 (11 + 3).; Writing 11 (the total).

S2. Each shape is a number. Find the number for the circle.

Answer: 6

The total is 10 and the square is 4. 10 - 4 = 6, so the circle is 6. Check: 6 + 4 = 10.

Watch for: Writing 14 (adding 10 and 4).; Writing 10 (the total).

S3. Each shape is a number. Find the number for the heart.

Answer: 8

8 + 8 = 16, so each heart is 8.

Watch for: Writing 16 (the total).; Writing 32 (adding the total twice).

S4. Each shape is a number. Find the number for the star.

Answer: 5

5 + 5 + 5 = 15, so each star is 5.

Watch for: Writing 3 (counting the stars).; Writing 12 (15 - 3).

S5. Each shape is a number. The square is 4. Fill in the blanks.

Answer: 10; 5

14 - 4 = 10, so the 2 triangles make 10. The triangles are the same, so each is 5. Check: 5 + 5 + 4 = 14.

Watch for: Writing 14 for each triangle.; Writing 10 for each triangle.

S6. Ben says the circle is 7. Look at the picture.

Answer: No; 6

7 + 7 = 14, but the line says 12. So Ben is not correct. 6 + 6 = 12, so the circle is 6.

Watch for: Saying Yes because 7 is close to 6.; Writing 12 for the circle.

S7. Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.

Answer: 10; 5

13 - 3 = 10 sweets are in the 2 bags. 5 + 5 = 10, so each bag has 5 sweets. Check: 5 + 5 + 3 = 13.

Watch for: Writing 13 for each bag.; Writing 6 (13 shared between 2 bags, ignoring the loose sweets).

Think harder (page 3)

T1. Each shape is a number. Find the number for each shape. Show your working.

Answer: square 6; heart 7

Line 1 has one more square than line 2, and 19 is 6 more than 13, so the square is 6. Then 6 + heart = 13, so the heart is 7. Check: 6 + 6 + 7 = 19.

Watch for: Writing 19 and 13 as the answers.; Finding the square but forgetting the heart.

T2. Kumar says the heart is 8 and the star is 7. Look at the picture. Decide if each statement is True or False.

Answer: True; False

Line 1: 8 + 7 = 15, which matches. Line 2: 8 + 8 + 7 = 23, but the line says 21. Kumar's numbers work in line 1 only.

Watch for: Checking only line 1.; Saying line 2 is true because 23 is close to 21.

T3. Each shape is a number. Find the number for each shape.

Answer: heart 3; circle 5

The 2 hearts make 6, so the heart is 3. Then 3 × circle = 15 means 3 groups of 5 make 15, so the circle is 5. Check: 3 × 5 = 15.

Watch for: Writing 6 for the heart.; Writing 15 for the circle.

Go further (page 4)

G1. Each shape is a number. Find the number for each shape.

Answer: circle 5; heart 7; star 11

Circle × circle = 25, so the circle is 5 (5 groups of 5). Then heart + 5 = 12, so the heart is 7. Then star - 7 = 4, so the star is 11. Check: 11 - 7 = 4.

Watch for: Writing 12 for the heart.; Taking 4 away from 7 and writing 3 for the star.

G2. Each shape is a number. Fill in the blanks.

Answer: 14; star 8; heart 6

Both lines together have 3 stars and 3 hearts, and 22 + 20 = 42. 3 stars and 3 hearts make 42, so one star and one heart make 14. Then 22 - 14 = 8 is a star, and 14 - 8 = 6 is a heart.

Watch for: Writing 22 and 20 as the answers.; Writing 7 for both shapes.

G3. Each shape is a number. The heart is greater than the triangle. Find the number for each shape.

Answer: heart 5; triangle 4

The pairs that add to 9 are 1 and 8, 2 and 7, 3 and 6, 4 and 5. Only 4 and 5 multiply to 20. The heart is greater, so the heart is 5 and the triangle is 4.

Watch for: Writing 4 for the heart.; Using 3 and 6 (they add to 9 but multiply to 18).

Investigation (page 5): One line or two?

Only circle 4 and triangle 6 fit both lines. One line alone has many pairs that fit (1 and 9, 2 and 8, 3 and 7, 4 and 6, 5 and 5 and more; accept 5 and 5 for line 1). 5 and 5 fails line 2 (5 + 5 + 5 = 15, not 14). The second line is the same as the first with one more circle, and 14 is 4 more than 10, so the circle is 4.

Look for: A child who tries pairs for both lines, finds that only one pair works, or says one line is not enough to find two numbers. Credit any clear reason. Do not mark a child down for stopping at one pair.

Exit ticket (page 5)

E1. Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.

Answer: 9

9 + 9 = 18, so each triangle is 9.

Watch for: Writing 18 (the total).; Writing 2 (counting the triangles).

E2. The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.

Answer: (4) 8

13 - 5 = 8, so the heart is 8. Check: 8 + 5 = 13.

Watch for: Choosing (1) 5, the number of the square.; Choosing (2) 6 or (3) 7 after a counting slip.

E3. Each shape is a number. The star is 5. Find the number for each circle.

Answer: 7

19 - 5 = 14, so the 2 circles make 14. 7 + 7 = 14, so each circle is 7. Check: 7 + 7 + 5 = 19.

Watch for: Writing 14 for each circle.; Writing 19 for each circle.

Sneak-peek puzzles (not in the workbook)

1. 5
2. 3. The two stars make 14, so one star is 7. The heart and 7 make 10, so the heart is 3.